JOHNS HOPKINS
BLOOMBERG SCHOOL
of PUBLIC HEALTH

Outcomes in Clinical Trials

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Outline

- Outcomes
  - Types
  - Influence on study design
  - Composite outcomes and surrogate outcomes

- Generalizability (external validity)

Introduction

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Definition

- ***Outcome*** and ***endpoint*** are often synonymous

- Typically, a quantitative measure
  - Defined at the study start
    - In the protocol and the statistical analysis plan (SAP)
  - Reflects the objectives of a trial
    - Efficacy
    - Effectiveness
    - Safety
  - Measured for each participant
    - Some aggregate of outcomes is analyzed per SAP and constitutes the result of a trial

Hierarchy of Outcomes—1

Primary outcomes

- Design variable
  - Used for determining or justifying sample size in planning a trial
- Related to the type of research
- Informed by the primary hypothesis

Hierarchy of Outcomes—2

Primary outcomes	Secondary outcomes

▶ Design variable
  ▶ Used for determining or justifying sample size in planning a trial
 	
▶ Related to the type of research
 	
▶ Informed by the primary hypothesis
	▶ Other important treatment effects
	  ▶ Safety
	  ▶ Mechanism of action
	
	▶ Allow for a more comprehensive evaluation of risk-benefit

Hierarchy of Outcomes—3

Exploratory outcomes

- Data on participant's health (other than primary or secondary outcome)
- Study participation
- Secondary effects of an intervention
- Hypothesis-generating

Hierarchy of Outcomes—4

- Typically, clinical trials are not powered to examine secondary or exploratory outcomes
  - Effects might still be seen, but if not—perhaps the statistical power is low

Criteria for Primary Outcome

- Relevant and important
- Likely to be influenced by the intervention
- Evaluable in all participants
- Assessment independent of treatment assignment (bias prevention)
- Power considerations (variability, frequency, anticipated differences)
- Statistical analysis plan (SAP)

Examples of Primary Outcomes

- Trial of a yeast versus placebo for the prevention of antibiotic-associated diarrhea
  - Proportion of participants developing diarrhea under antibiotic treatment

- Trial of two methods of in vitro fertilization (comparative effectiveness)
  - Proportion of participants experiencing preeclampsia/eclampsia

- Trial of a drug versus placebo to treat agitation in Alzheimer's disease
  - Proportion of participants with a reduction in agitation

Types of Outcomes

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Event Outcomes—1

- 0/1 value for absence/presence; no/yes; normal/abnormal
  - Clinical event
    - E.g., myocardial infarction (MI), stroke, death
  - Cut-off value
    - E.g., anemia, hypertension

Event Outcomes—2

- Timing
  - Measured at a fixed timepoint
    - E.g., coronary heart disease (CHD) at two years
    - Missing data can be an issue
  - Time to event
    - E.g., time to diagnosis of CHD
    - Allows for censoring
    - More powerful than fixed-time analysis

Ordinal Outcomes

- Ranked categories (1–5; A–D)
- Difference between categories is usually qualitative, not quantitative
  - Staging systems, adverse event severity
- Analysis methods
  - Proportional odds model, cumulative logit model
  - Chi-square test for trend
- Ignoring the inherent scale in the analysis decreases power

Continuous Outcomes

- Outcome is a value
  - Standard units
    - E.g., lab values
  - Need to define a clinically important difference
- Often more powerful and easier to interpret than other outcomes
- Often measured as a change from baseline
  - I.e., “control” for initial value
- Distributional assumptions may be important

Objective Versus Subjective—1

Objective: clinical events or measurements that are definitive and require little (or no) judgment

- Rigorous definitions to limit interpretation (subjectivity)
- May include test results for confirmation

Objective Versus Subjective—2

Subjective: assessments that rely on judgments

- Histology, expert assessment, need for medication
- Health status or change in status
  - E.g., pain relief, quality of life (QoL)
- Masking is important, if possible!

Enhancing Quality and Accuracy of Outcome Ascertainment

- Define criteria for evaluation
- Pretest and validate measurements
- Train and (if possible) mask evaluators
- Panel adjudication, locally or centrally
  - Independent assessments
  - Carefully define the process
  - Ongoing quality assurance
    - Standards
    - Replicate measurements

Influence of Outcomes on Study Design

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Influence on Study Design—1

- Eligibility criteria
- Sample size
- Length and frequency of follow-up
- Need for and type of masking

Influence on Study Design—2

- Personnel and resources required
- Quality assurance required
- Costs

Constructing Outcomes—“Three Bs”

Do your best to balance the following:

Biology

- Does the outcome reflect a clinically relevant fact/change?

Biostatistics

- Detectable difference between groups is plausible and practical

Budget

- Must be able to afford to measure reliably for every participant

Composite Outcomes and Surrogate Outcomes

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Composite Outcomes

- Two or more events related to disease progress capture different aspects of response
- Example: outcome occurs if participant suffers cardiovascular death, myocardial infarction (MI), or stroke

Why Use Composite Outcomes?

- More events, reduces sample size
- At times, there is no obvious single primary outcome
- May be a measure for a pathophysiological mechanism

Composite Outcomes: Issues

- Can be driven by one frequent single endpoint (part of the composite)
- Often, single endpoints are not of similar importance
- Effect size for the single endpoints might differ
- Can be hard to interpret

Thoughts on “Quality Criteria” for Composite Outcomes

- Are the single endpoints of similar importance?
- Is the incidence of the single endpoints similar?
- Is the effect for the single endpoints similar?
- (Do the single endpoints share a common underlying mechanism?)

Source: McCoy C. E. (2018). Understanding the use of composite endpoints in clinical trials. *The Western Journal of Emergency Medicine*, *19*(4), 631–634. https://doi.org/10.5811/westjem.2018.4.38383

Surrogate Outcomes

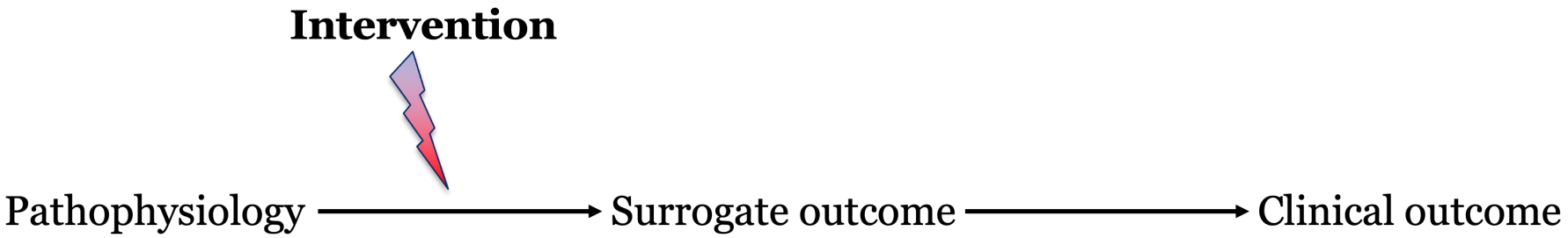
"A laboratory measurement or physical sign used as a **substitute** for a clinically meaningful endpoint that measures directly how the patient feels, functions or survives. Changes induced by a therapy on a surrogate endpoint are expected to reflect changes in a clinically meaningful endpoint."

Source: US Food and Drug Administration. (n.d.). *Clinical trial endpoints.* Eugene J. Sullivan, MD, FCCP; Principal, EJS Consulting, LLC. Accessed on November 30, 2021, from https://www.fda.gov/media/84987/download

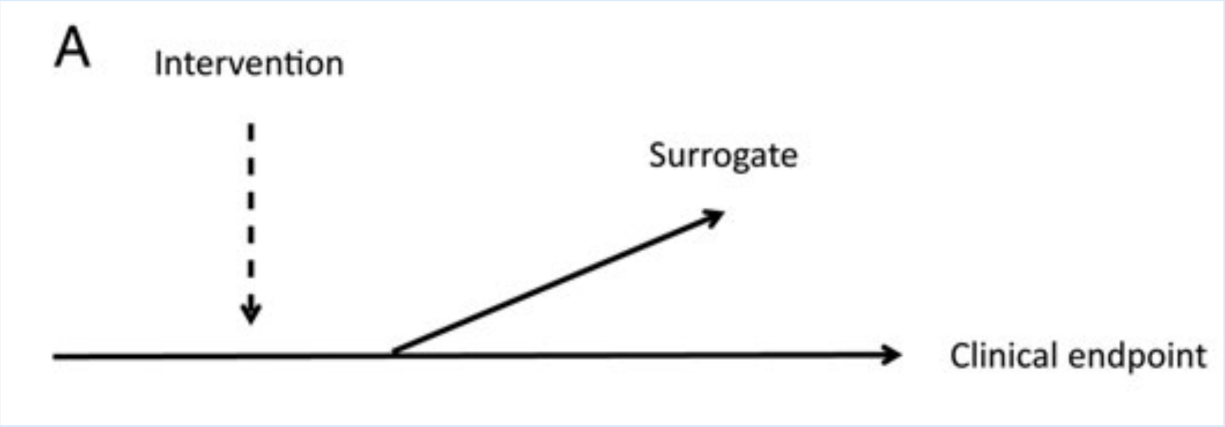
Surrogate Outcomes—Examples

Surrogate outcome		Clinical outcome
Cholesterol	→	MI
Ventricular arrhythmias	→	Sudden cardiac arrest
CD4+ T-cells	→	AIDS
Arterial occlusions	→	Stroke, cardiac events

Model for Surrogate Outcomes

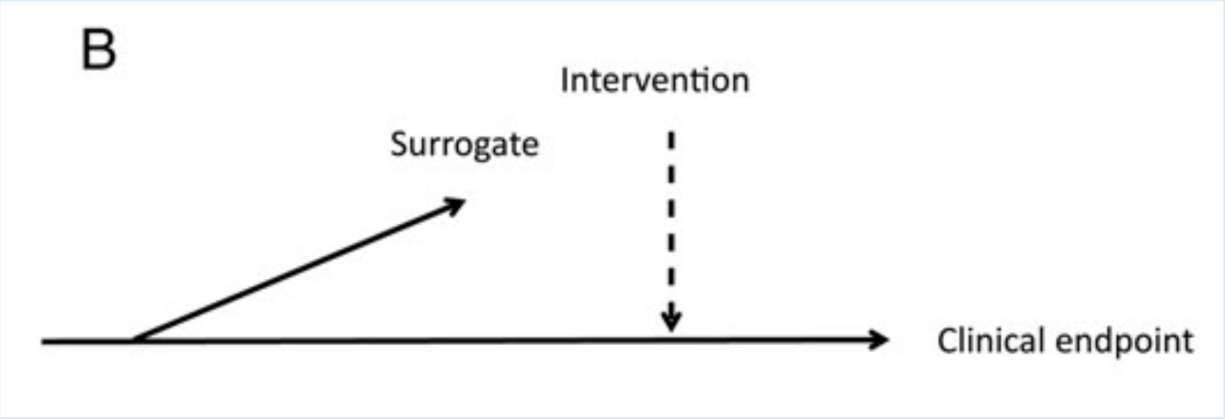


Model A for Surrogate Outcome



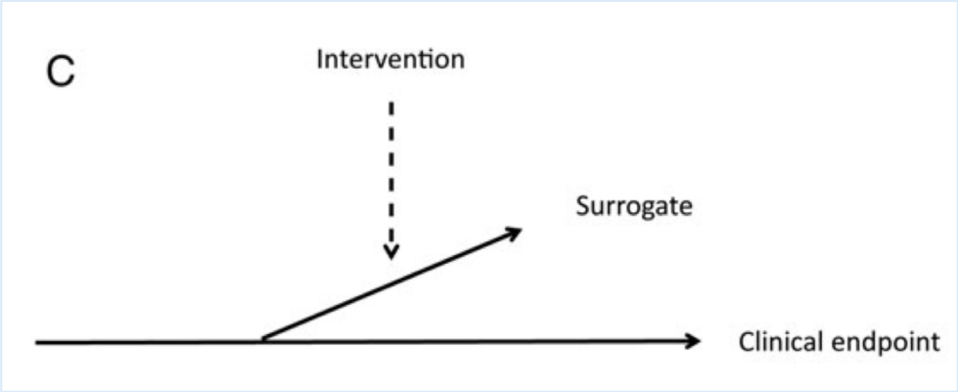
Source: Weintraub, W. S., Lüscher, T. F., & Pocock, S. (2015). The perils of surrogate endpoints. *European Heart Journal*, *36*(33), 2212–2218. https://doi.org/10.1093/eurheartj/ehv164

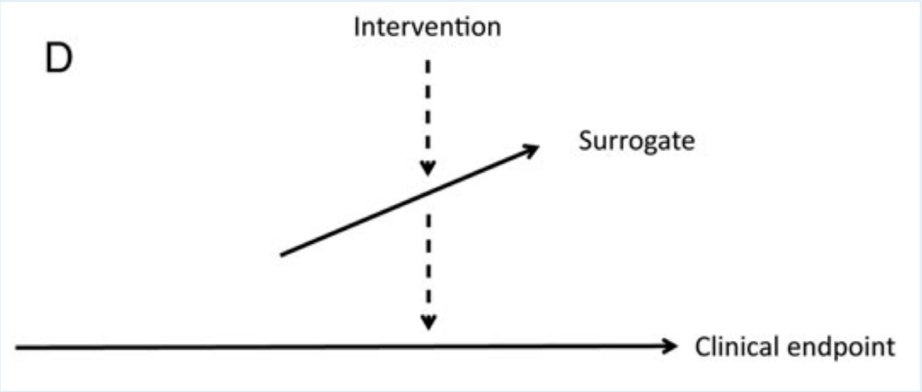
Model B for Surrogate Outcome



Source: Weintraub, W. S., Lüscher, T. F., & Pocock, S. (2015). The perils of surrogate endpoints. *European Heart Journal*, *36*(33), 2212–2218. https://doi.org/10.1093/eurheartj/ehv164

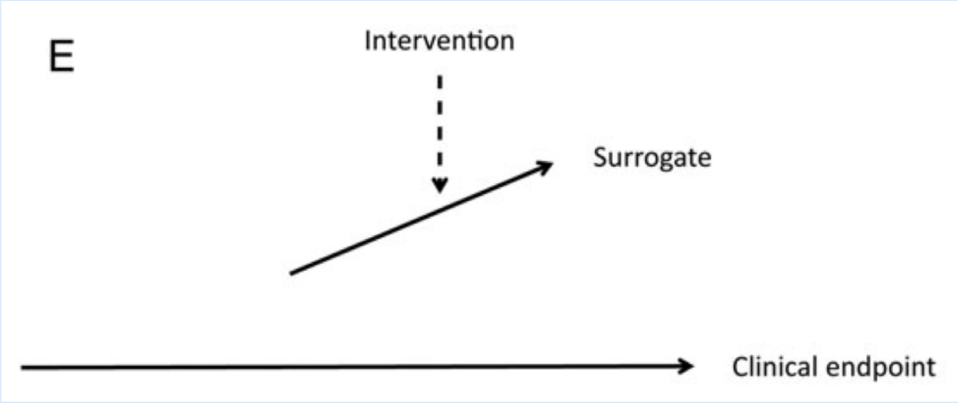
Models C and D for Surrogate Outcome

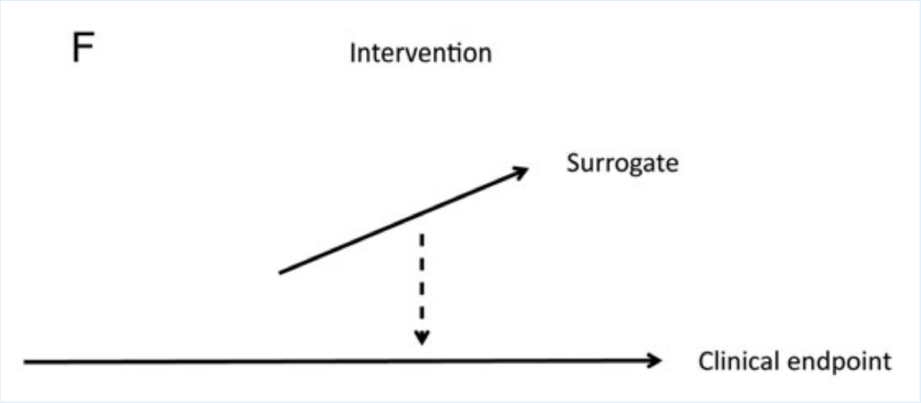




Source: Weintraub, W. S., Lüscher, T. F., & Pocock, S. (2015). The perils of surrogate endpoints. *European Heart Journal*, *36*(33), 2212–2218. https://doi.org/10.1093/eurheartj/ehv164

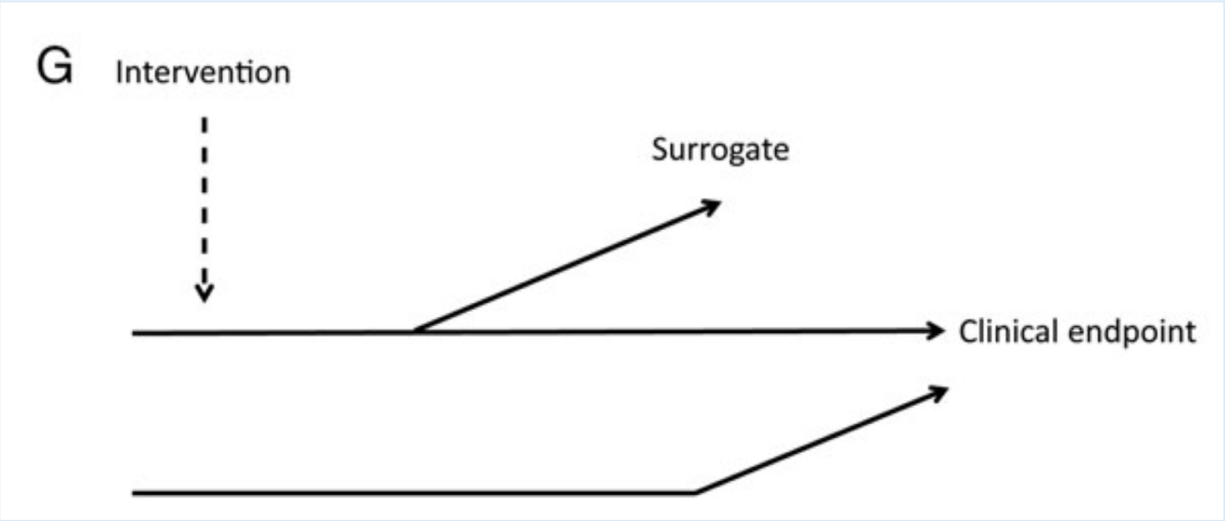
Models E and F for Surrogate Outcome





Source: Weintraub, W. S., Lüscher, T. F., & Pocock, S. (2015). The perils of surrogate endpoints. *European Heart Journal*, *36*(33), 2212–2218. https://doi.org/10.1093/eurheartj/ehv164

Model G for Surrogate Outcome



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Why Use Surrogate Outcomes?

- Reduced sample size
  - More frequent events
  - Continuous rather than binary variables
- Shorter follow-up
  - E.g., weeks versus years
- Faster drug development
  - Reduce costs, faster benefit for patients
- May be a more direct measure of efficacy
  - Less affected by competing risks, other treatments

Example

Primary outcome	Number of participants	Length of follow-up
Mortality, fatal coronary event or MI, or first acute coronary event	4,200–6,600	5 years
Plasma lipoproteins	75–100	3–6 months

Evaluating Surrogate Outcomes

- Define patients, treatments, and clinical endpoints for which the potential surrogate applies
- A strong statistical association between the surrogate and the clinical outcome of interest
- Strong, consistent evidence of treatment differences in the surrogate
- Treatment difference in clinical outcome within each trial is statistically explained by the surrogate
- Across trials, magnitudes of treatment difference in the surrogate and in the clinical outcome are closely linked

Source: Weintraub, W. S., Lüscher, T. F., & Pocock, S. (2015). The perils of surrogate endpoints. *European Heart Journal*, *36*(33), 2212–2218. https://doi.org/10.1093/eurheartj/ehv164

Generalizability (External Validity)

Generalizability (External Validity) Versus Internal Validity

Internal validity	External validity/generalizability
▶ Inference from experiment is free from bias and confounding ▶ Conclusions regarding the study population are correct ▶ "Trial done and analyzed correctly"	▶ Degree to which the results of the study may apply, be relevant, or be generalized to populations or groups that did not participate in the trial

Another Definition of Generalizability (External Validity)

- Extent to which results of a trial provide the correct basis for generalization to other circumstances
  - Settings
  - Persons
  - Treatment regimens
  - Designs
  - Outcomes
  - Follow-up

Source: Juni, P., Altman, D. G., Egger, M. (2001, July 7). Systematic reviews in health care: Assessing the quality of controlled clinical trials. *British Medical Journal, 323,* 42–46. Adapted from Horton, R. (2000, December 15*). Statistics in Medicine, 19*(23), 3149–3164.

Why Is Generalizability (External Validity) Important?

- We want to apply our trial findings to groups other than the population we studied
- Can’t redo every trial across different unique settings
- Serve the health of the public

Summary

- The careful choice of outcomes is critical for study design, implementation, analysis, interpretation, and relevance
- Composite outcomes can decrease sample size but need to be constructed with care to avoid methodological difficulties and unclear interpretation
- Surrogate outcomes may substitute for a clinically meaningful measure to shorten the duration of a trial
- A trial should be developed with generalizability aspects in mind